MODE-S Wind on a phone

The mobile view `/app`: Skew-T, time–height section, ILS approach and approach conditions in your pocket

Otso Laakso (OH2GAX)
Tuusula, Finland · near Helsinki-Vantaa airport (EFHK)

In short. The full MODE-S Wind pages are made for a computer screen. The mobile view is a light page for phones: open the address of MODE-S Wind with `/app` at the end. It shows *one product at a time*, filling the screen:

- **Skew-T · Area now** — the temperature and wind profile measured by the aircraft over the last 30 minutes to 3 hours, with the radiosonde and the MEPS model if you like;
- **Skew-T · Area hour** — any stored hourly profile;
- **Time–height** — the hourly profiles side by side: fronts, inversions and jet streams as they pass;
- **ILS approach** — the aircraft on final on the 3° glideslope and on a map, live, with the wind measured along the approach, any windshear alert and, if you like, the ACI + windshear chart;
- **ACI · WS · ST** — the approach conditions index and windshear of every landing and departure (also with the turbulence, EDR), the space–time diagram of the wind changes, or the boundary layer (low-level jet, inversion), with the flights of the period as a list;
- **History** — the landings or the departures of 1 hour to a day, or of any date, as cards with their winds per height band and their reports, with a search; and the METARs and pilot reports;
- **Statistics** — what the receiver hears: message rate, downlink formats, meteo yield, range rings and more, the sections of your choice.

Everything is chosen from the ≡ menu at the top left. Tap a diagram for its values. *Add to Home Screen* gives MODE-S Wind its own icon, and it opens full screen like an app. The diagrams are the same as on the full site, so the other user guides explain how to read them; this guide explains how to use them on a phone.

Contents

1 Getting started **2**

2 The menu **2**

3 Touch, rotation and updates **4**

4 Skew-T: Area now and Area hour **5**

5 Time–height **7**

6 ILS approach **8**

7 ACI · WS · ST **13**

8 History **18**

9 Statistics **22**

10 When something looks wrong **23**

11 Quick reference **24**

1 Getting started

1.1 Opening the mobile view

Open the phone's browser and go to the address you use for MODE-S Wind on a computer, with `/app` added at the end — for example `http://your-address/app`. The phone has to reach MODE-S Wind the same way a computer does (your own network, or the connection set up for it). The browser asks for the same user name and password as the other pages and normally remembers them.

The mobile view opens the product you looked at last. All choices — the view, its options, the theme — are remembered on the phone. The Skew-T options and a chosen Light or Dark theme are the same settings as on the Live Map and the Sounding page *of the same browser*: if you also open the full site on the phone, it starts with the same choices (the full site has no Auto).

1.2 Add to Home Screen

Android (Chrome): open `/app`, tap the browser menu ⋮ and choose Add to Home screen (on some phones Install app), then Add.

iPhone and iPad (Safari): open `/app`, tap the Share button (the square with an arrow) and choose Add to Home Screen, then Add.

The MODE-S Wind icon (right) appears among your apps. It opens the mobile view full screen, without the browser's address bar, and can be switched to like any other app. The first time it is opened the phone may ask for the user name and password once more. From 28 October 2026 the login is then **remembered for 30 days** (renewed while you use MODE-S Wind) — also after the phone's screen has been off or the browser has been closed.



the icon

1.3 The screen

A slim bar at the top holds five things (Fig. 1):

- ≡ at the left opens the menu (Section 2);
- the **name of the view** and under it a short line: what is shown and when it was updated — e.g. `last 60 min · 15:55 UTC`, `24.10.2026 15–16 UTC`, or on the ILS view the number of aircraft on the approach, the QNH and any active windshear alert;
- the **current time in UTC** (from 28 October 2026), hours and minutes — all times of MODE-S Wind are UTC, two hours behind Finnish winter time and three behind summer time;
- the **theme button** (from 28 October 2026): *moon* — dark theme, *sun* — light theme, *half-filled circle* — **Auto**: the theme follows the phone's own dark / light setting, and changes with it while the page is open (e.g. when the phone turns dark in the evening). Each tap steps *Dark* → *Light* → *Auto* → *Dark*; a short note under the button names the new setting. Auto is the setting until you choose one; your choice is remembered on the phone;
- a **dot** at the right: the system health of MODE-S Wind, the same indicators as on the Statistics page. *Green*: all parts work. *Amber*: something is late, retrying or using a fallback (e.g. no new MEPS model data). *Red*: a part has stopped. The menu's General section says which part; the Statistics page on the full site has the details.

Below the bar is the product. All times are UTC.

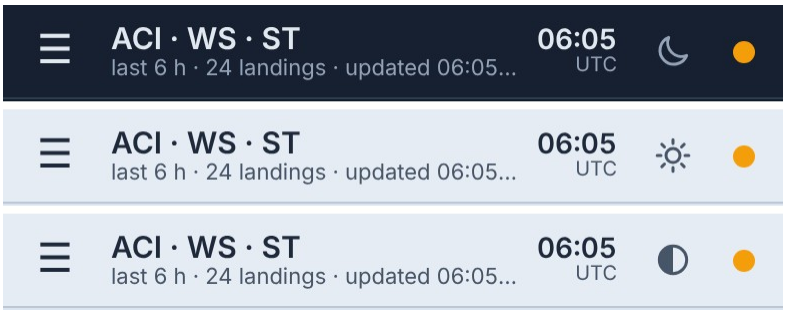


Figure 1. The top bar with the theme set to Dark (moon), Light (sun) and Auto (half circle; here the phone is set to light): the view and its status line, the current UTC time, the theme button and the system-health dot.

2 The menu

Tap ≡. The menu slides in from the left (Fig. 2); × or a tap beside it closes it.

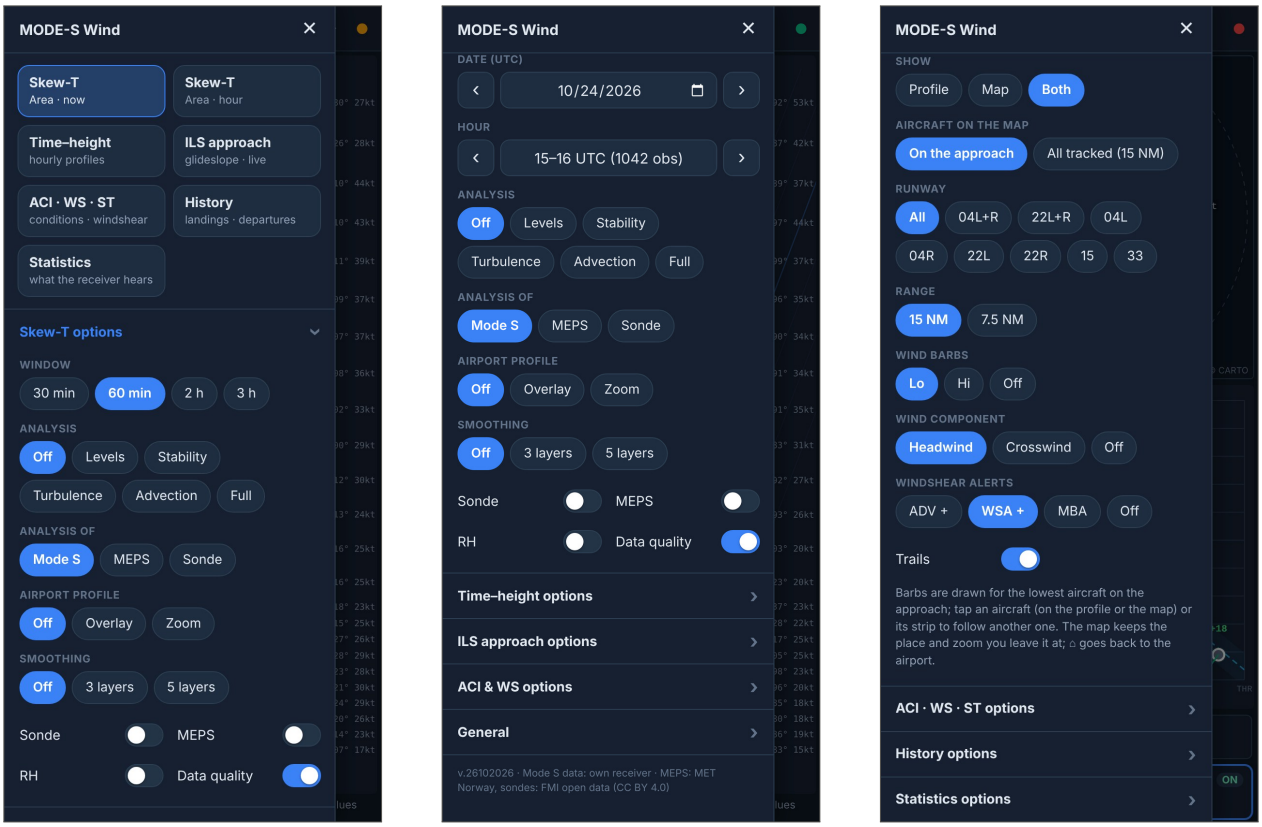


Figure 2. The menu. Left: the seven views as tiles (the shown one highlighted) and the options of the shown view, here Skew-T, opened below them. Middle: Area · hour with its date and hour steppers and the switches. Right: the ILS approach options (from 28 October 2026 with Show and Aircraft on the map); the other products' options are folded but one tap away.

Views. The seven tiles at the top choose the product. A tap switches to it and closes the menu.

Options. Below the tiles each product has a folding section: Skew-T options, Time–height options, ILS approach options, ACI · WS · ST options, History options and Statistics options. The shown view's section opens by itself (its title in blue); the others stay folded and open with a tap, so you can, for example, set the Skew-T options while the History is shown. The menu scrolls on its own.

Controls. Choices are buttons: the highlighted one is in use, a tap on another changes it at once. On / off settings are switches. Dates have a calendar field and, for stored hours, ⟨ ⟩ buttons that step one day or one hour back and forward (over midnight to the next or previous day). Changes take effect immediately; there is no *Apply* button.

General. The system health in words, What's new, About and User guides (from 28 October 2026, Fig. 3), a link to the full site and to this guide. What's new shows the same short notes of each new version as the full site's *What's new* window — the newest one open, the older ones folded (tap to open). What's new is also at the top of the menu, next to the *MODE-S Wind* title (from 30 October 2026), so it is easy to find when there is something new. At the top of What's new there is a User guides button, and at the top of User guides a What's new button, so you can move between the two directly. A small **red dot** on ≡ and on both buttons tells that there is a version you have not read yet; it goes away when you open What's new (also when you have read it on the full site in the same browser). About shows the same text as the About window of the full site: who made MODE-S Wind and how to reach him, the receiver site, how the observations are made, the products, the windshear detection and the credits. User guides lists all the guides of the full site — e.g. the Windshear guide or the windshear study log — with their version, date, size and contents (Contents); Open PDF opens one in the phone's PDF viewer, Download saves it. Both open over the view; × or the phone's *Back* closes them.

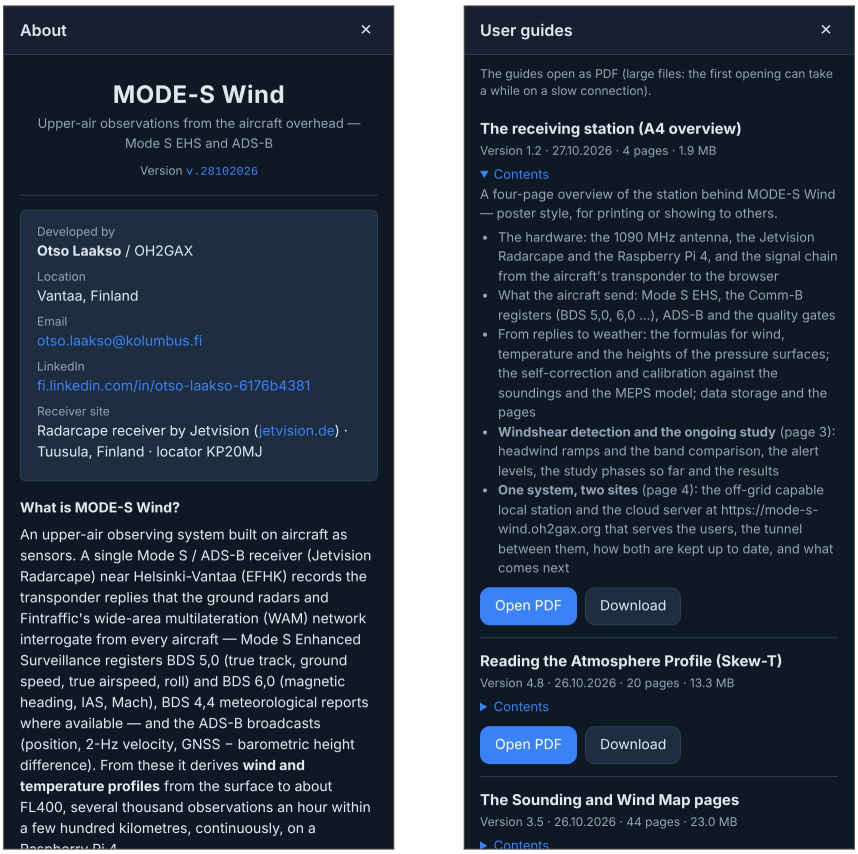


Figure 3. About (left) and User guides (right, with the contents of the first guide opened), from the menu's General section.

3 Touch, rotation and updates

Tap for values. On the full site the values appear when the mouse is over a diagram. On the phone, *tap* the diagram: a box shows the values at that point — on the Skew-T the level's temperature and wind (and the sonde's and model's), on the time–height section the hour and layer, on the ACI & WS chart the nearest landing, departure or go-around, on the space–time diagram the flights of that cell. *Drag* sideways or up and down to move along; tap elsewhere to move the box. On the ILS view a tap on an aircraft chooses whose wind barbs are drawn (Section 6).

Scrolling. When a view is taller than the screen — ACI · WS · ST with its list of flights, the History, the ILS view with many aircraft, or a Skew-T turned to landscape — swipe up and down to scroll, also when the swipe starts on the diagram. A short tap still gives the values.

Portrait and landscape. In portrait each diagram takes the full width and as much of the height as it needs. Turn the phone for more room across: the time–height section gets a longer time axis, the ILS profile spreads over the screen (Fig. 4); the Skew-T keeps its shape and the page scrolls.

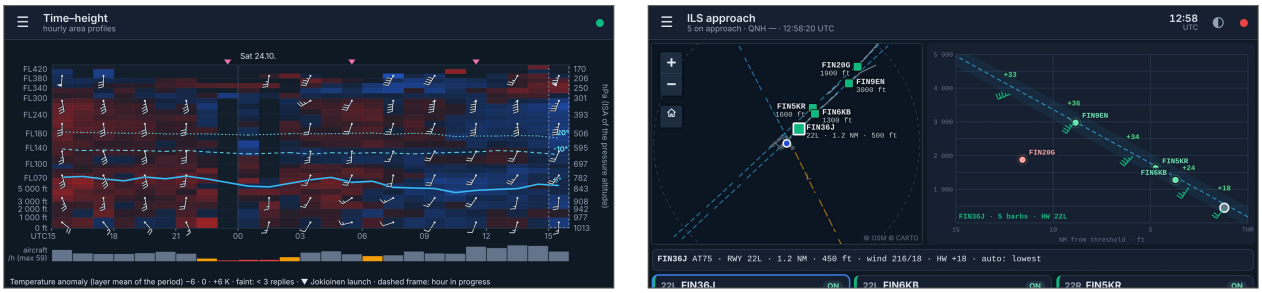


Figure 4. Landscape: the time–height section (24 hours) and the ILS approach with Show: Both — the map beside the profile. The strips of the ILS view are below; scroll down to see them. (The ILS picture: approach traffic replayed from the stored observations; the map tiles are not shown, see Fig. 11.)

Updates. Only the view on the screen is updated, and nothing while the phone's screen is off or another app is in front; when you come back the view is updated at once. Table 1 gives the intervals. The ILS view is the only one that updates every few seconds: it loads only the aircraft on the approach and the active alerts, compressed — about 1–5 kB each

time, depending on the traffic — so watching it for an hour on mobile data uses roughly 1–4 megabytes. Its map adds the aircraft positions (well under 1 kB each time) and, once, the map program and the map tiles (about half a megabyte; the phone keeps them). The Statistics view uses about 7 megabytes an hour (Section 9); the other views use little.

View	Updated	Data
Skew-T · Area now	every 2 min	the area profile; METAR every 10 min
Skew-T · Area hour	when you choose an hour	stored hourly profile
Time–height	every 10 min	hourly profiles of 12 h – 14 days
ILS approach	every 5 s	live approach traffic (with the map: their positions), alerts; METAR and QNH every 10 min
ACI · WS · ST	every minute	landings, departures, go-arounds of 3 h – 1 day; space–time diagram
History	every minute (not for a date)	landings or departures of 1 h – 1 day or of one date
Statistics	every 30 s	the Statistics page's data of the window (about 55 kB)
Health dot	every minute	the system health indicators

Table 1. When the views are updated (only while shown and the screen is on).

4 Skew-T: Area now and Area hour

The Skew-T is the same diagram as the Live Map's *Atmosphere Profile* and the Sounding page (Fig. 5): temperature against pressure, with the isotherms tilted to the right, and the wind as barbs at the right. The grey curve and the grey barbs with their values are the *aircraft* — the average of all aircraft within 150 km. With Sonde the Jokioinen radiosonde is drawn in pink (barbs inside the right edge of the plot), with MEPS the MEPS weather model in teal (barbs in the margin, just left of the aircraft's). The labels at the top left give the data quality, the sounding's and the model's time and how far the aircraft are from them.



Figure 5. Left: Skew-T · Area now, the last 60 minutes, with Sonde (pink) and MEPS (teal) on; the labels give how far the aircraft are from them: on average 0.0 K from the 12 UTC sounding and −0.3 K from the model, wind 10 and 7 kt RMS. Right: Area · hour 24 October 2026 15–16 UTC with Analysis: Levels (tropopause, 0 °C level, maximum wind); a tap at 447 hPa shows the aircraft (−29.7 °C, 191°/35 kt), the sonde and the model at that level.

Area now is the live profile, updated every 2 minutes, with the latest METAR at the ground (ground line, temperature and dewpoint). **Area hour** shows one stored hour: choose the date and the hour in the menu, or step with ⟨ ⟩. The METAR of that hour is stored with it.

Option	Choices	What it does
Window	30 min, 60 min, 2 h, 3 h	period of Area now; longer = more aircraft, less up to date
Date, Hour	calendar, ⟨ ⟩	the stored hour of Area hour
Analysis	Off, Levels, Stability, Turbulence, Advection, Full	automatic marks: tropopause, isotherms, jets, inversions, stability, turbulence risk, warm / cold advection, contrails, surface parcel
Analysis of	Mode S, MEPS, Sonde	which profile the analysis is computed from; MEPS and Sonde add likely cloud layers
Airport profile	Off, Overlay, Zoom	the fine low-level profile of the landings and departures at EFHK; Zoom shows the lowest 5 000 ft; from 29 October 2026 with the hour's low-level jet (◊ LLJ) and surface inversion (red SFC INV bracket)
Smoothing	Off, 3 layers, 5 layers	vertical smoothing of the aircraft profile
Sonde	switch	the Jokioinen radiosonde
MEPS	switch	the MEPS weather model over the same area
RH	switch	the trial moisture strip with the sonde's and the model's humidity
Data quality	switch	the data badge and the rings at layers with only 1–2 aircraft

Table 2. Skew-T options. The *Reading the Atmosphere Profile (Skew-T)* guide explains each of them.

5 Time–height

The time–height section puts the hourly profiles side by side: time runs to the right, height up (Fig. 6). The colour is the chosen field, the barbs the hourly wind; lines mark the 0, −10 and −20 °C levels, the tropopause and the maximum wind when switched on. The strip at the bottom gives the aircraft per hour; a pink triangle at the top marks a Jokioinen launch, and the last column (dashed frame) is the hour in progress. Tap a cell for its values.

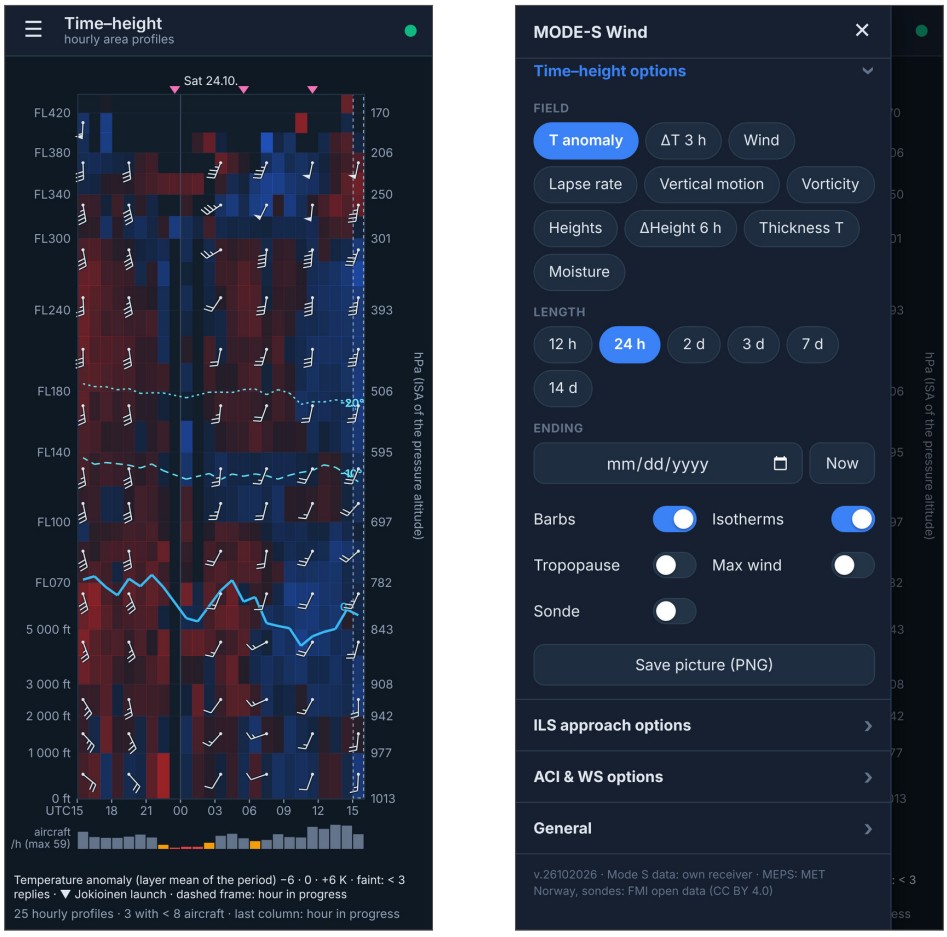


Figure 6. Time–height, 24 hours to 24 October 2026 15 UTC, temperature anomaly (blue colder, red warmer than the period's mean in that layer), with barbs and isotherms; right: its options.

Options. Field: temperature anomaly, temperature change over 3 h, wind speed, lapse rate, vertical motion, vorticity, heights of the pressure surfaces, height change over 6 h, thickness temperature check, the moisture trial and the *boundary layer* — BL wind and BL temp, the lowest 5 000 ft at the airport on a linear scale (Fig. 7). Length: 12 h to 14 days — 24 h by default on a phone; landscape gives the longer periods room. Ending: a past date, or Now. Switches for barbs, isotherms, tropopause, maximum wind, the sonde as its own column and the Boundary layer — the low-level jet (◊ with its speed in kt, joined hour to hour) and the surface inversion (red, from the ground to its top) on any field. Switches that do not apply to the BL fields are greyed out. Save picture (PNG) stores the section as an image on the phone. The *Sounding and Wind Map pages* guide explains the fields.

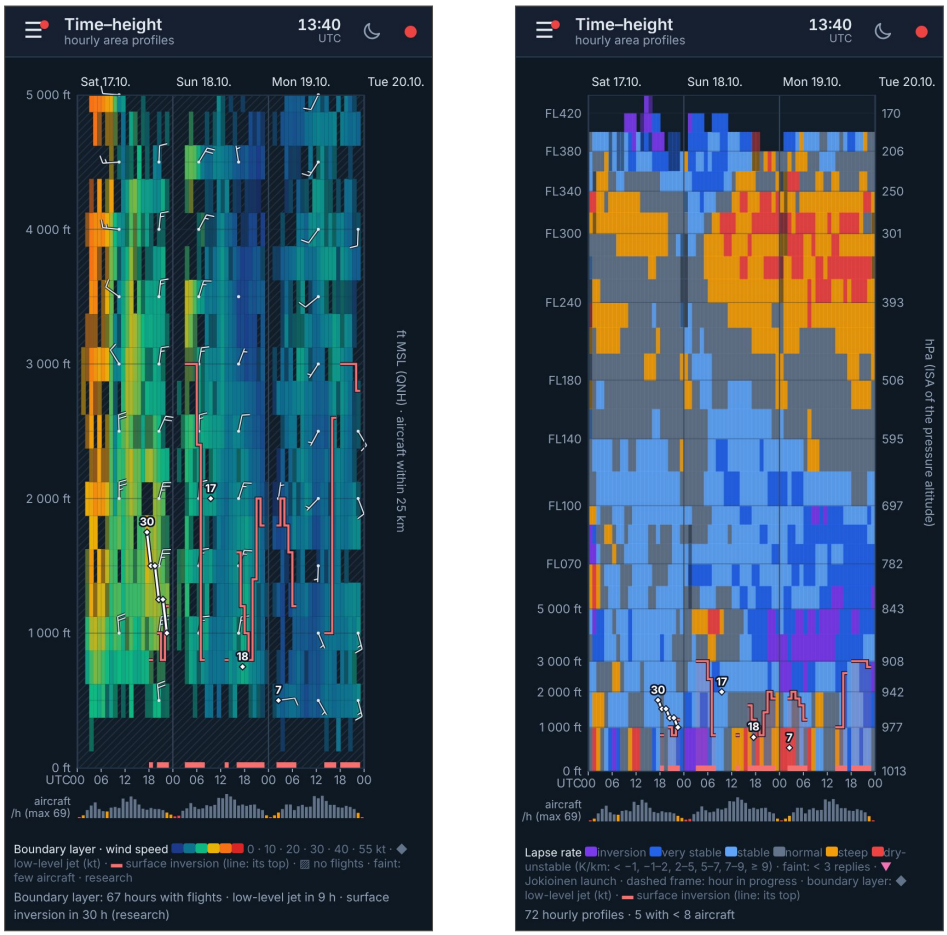


Figure 7. Left: BL wind, 72 hours to the end of 19 October 2026 — the wind of the aircraft within 25 km in 250-ft steps (blue calm, green, yellow, red 55 kt), the low-level jet of the evening of 17 October (30 kt at 1 750 ft, sinking) and the surface inversions under the high (red bar, red line at the top). Hatched: no flights. Right: Lapse rate with the Boundary layer switch on — the jet and the inversions in the lowest 3 000 ft.

6 ILS approach

The ILS view is the Windshear page's vertical profile of the final approach, live (Fig. 8): distance from the runway threshold to the right (15 NM to the threshold), altitude up (to 5 000 ft). It is redrawn every 5 seconds.

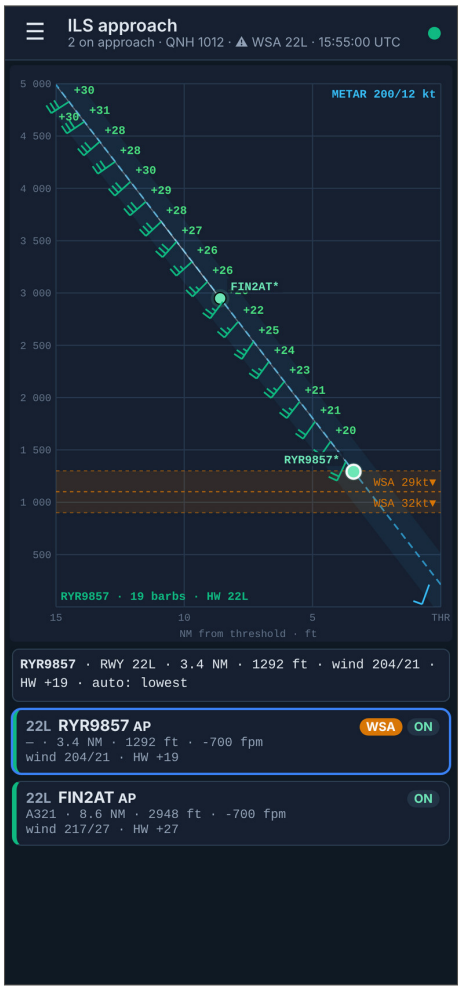


Figure 8. ILS approach, runway 22L. Two aircraft on the glideslope (green `ON`); the barbs are those of the lower one, RYR9857, measured along its approach from 15 NM in, with the headwind component of each in kt above it; a windshear alert (`WSA`) is active at 900–1 300 ft. The METAR surface wind is the blue barb at the threshold. Under the profile the chosen aircraft and the strips. *Illustration with simulated approach traffic.*

What is drawn.

- **The glideslope**: the dashed line is the 3° path (318.5 ft per NM above the threshold, corrected for the QNH), the shaded band ±300 ft around it.
- **Aircraft** in the approach corridors as dots with their callsigns (⋆: autopilot engaged): green *on* the glideslope, amber *high*, red *low*; the faint line behind each is its path over the last minute.
- **Wind barbs** of one aircraft, measured by it on the way down, coloured by where the wind comes from (green: computed from the aircraft's speed and heading replies; blue, orange, violet: other report types). The number above each barb is the headwind component for that runway (Headwind; positive = headwind) or the crosswind (Crosswind). Small hollow circles mark points where no wind could be computed (amber: the aircraft was turning; grey: a gap in its position data). The line at the bottom left names the aircraft and the number of barbs.
- **Windshear alerts**: an active alert is a band across the profile at its altitudes, blue `ADV` (advisory), amber `WSA` (windshear alert) or red `MBA` (microburst alert), with the change in kt (▼ loss, ▲ gain of headwind).
- **METAR**: the surface wind as a blue barb at the threshold, its text at the top right.

The map (from 28 October 2026). With Show: Both — the default — a map of the approaches is above the profile (beside it in landscape), with Map only the map, with Profile only the profile as before (Fig. 11). The map shows the aircraft as on a radar screen: an open square framed in the colour of the wind source (as the barbs) with the callsign and altitude; the aircraft whose barbs are drawn has its square framed thinly in white (in the light theme in black, which stands out on the light map), from 30 October 2026 the same size as the other approach traffic and with the same label (in the pictures below still larger).

Departures (from 29 October 2026). The climb-outs are not shown by default, as on the Windshear page. With Departures on the map: Shown they are drawn like the other traffic, with their runway and an arrow in the label, e.g. `A016 · 22R↑` — useful to see a departure climbing through the layers the arrivals descend in. A go-around is not a departure:

it stays with the approaches.

Boundary layer line (from 29 October 2026; research). Under the selected aircraft a line always tells what the lowest air is doing (Fig. 9): *Boundary layer, last hour* with a chip `LLJ 750 ft · 18 kt 023°` when there is a low-level jet and a red chip `surface INV +2 K` when the air at 600–1 000 ft is warmer than at the ground; otherwise grey *no jet · no inversion*, and *no recent flights* at night when nothing flew. A jet means the headwind changes on final below its height; an inversion means calm, smooth air near the ground (Section 7).

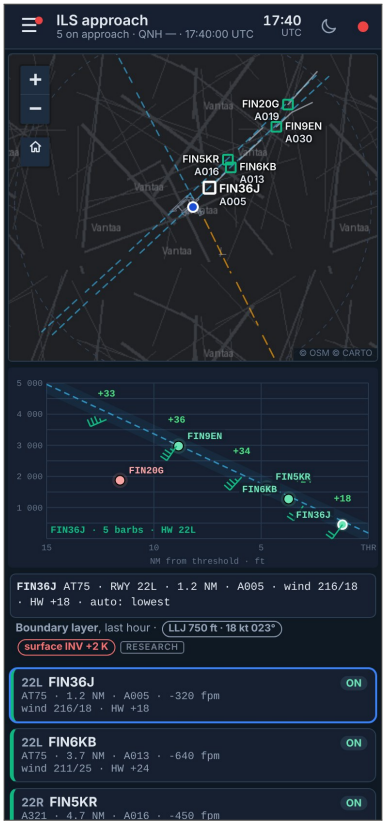


Figure 9. The boundary-layer line under the selected aircraft: a low-level jet at 750 ft and a surface inversion of +2 K (18 October 2026 17 UTC).

Turbulence on the map (from 29 October 2026; research). The option Turbulence on the map (EDR, research) (Off by default, 30 min, 1 h) colours the straight pieces (about 2 km each) of the paths of the landings of that period — and of the departures when Departures on the map is Shown (from 30 October 2026) — with thin lines by their turbulence, the EDR (Section 7): green smooth, yellow light (from 0.10), orange moderate (from 0.20), red severe (from 0.45). Where several flights flew the same path the strongest is drawn on top, so the colour shows the worst piece. A line under the selected aircraft sums it up, e.g. *Turbulence (EDR), last 60 min · 9 landings, 3 departures · max 0.26 moderate at 1400 ft (FIN8MC 22L) · median below 1000 ft 0.15, 1000–3000 ft 0.12* — the departures are counted only when they are shown (Fig. 10).

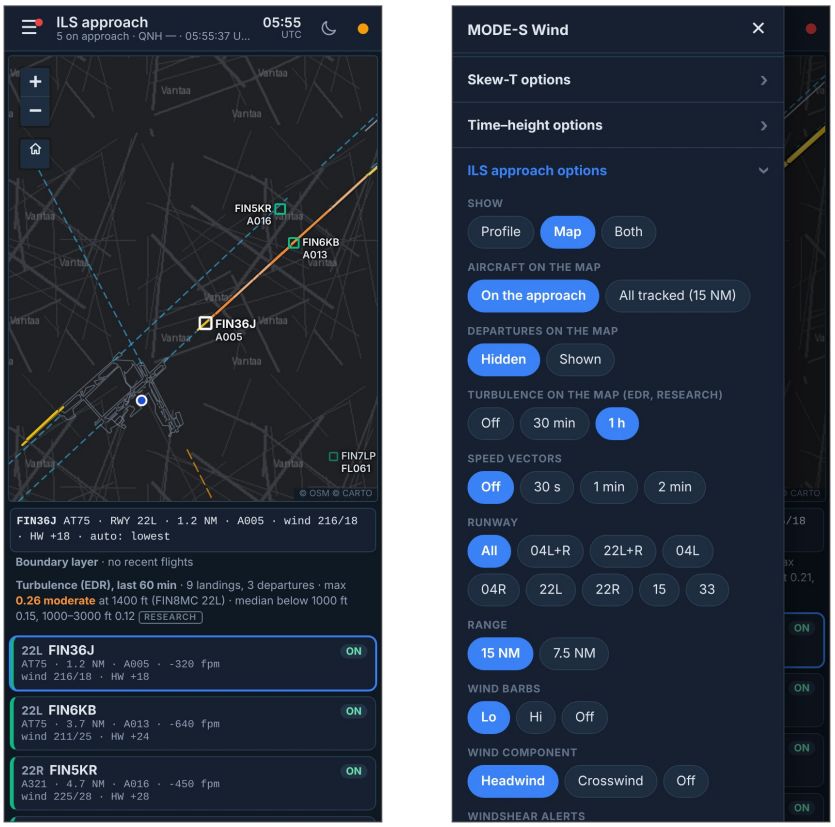


Figure 10. Left: Turbulence on the map 1 h with Departures on the map: Shown — the final approach to 22L in orange (moderate) and yellow (light), a departure at the lower left; the flights of 28 October 2026 replayed in a test. Right: the option in the menu.

Speed vectors (from 29 October 2026). The option Speed vectors (Off by default, 30 s, 1 min, 2 min) gives each aircraft a line to where it will be after that time at its ground speed along its track, as on a radar screen — in the colour of its square, white for the selected aircraft. A faster aircraft has a longer line: below 5 000 ft, at 140–250 kt, 30 s is 1.2–2.1 NM and 1 minute 2.3–4.2 NM, about the spacing between the aircraft on final. Start with 30 s; 1 min shows the spacing to the aircraft ahead. A landed aircraft gets none.

Altitudes as a controller sees them (from 29 October 2026). On the map, the strips and the line under the profile the altitude is written as on an air traffic controller's radar label: `A023` is 2 300 ft above sea level on the airport's QNH (the METAR pressure), and above the transition altitude of 5 000 ft the flight level, e.g. `FL061`. Aircraft transponders report their altitude on the standard pressure of 1013.25 hPa, so on a low-pressure day the raw value is several hundred feet too high: at QNH 1003 hPa an aircraft reporting 1 275 ft is really at about 1 000 ft, `A010`. The profile's height axis and the glideslope already use the QNH.

ATC labels (from 30 October 2026). The option Labels on the map: ATC (Standard by default) gives every aircraft on the map the three-line label of the ATC style of the Live Map and the Windshear page, in a small monospaced font: the callsign; the level in hundreds of feet — the altitude on the METAR QNH up to the transition altitude of 5 000 ft, the flight level above, without the letters — with ↑ or ↓ when climbing or descending at 300 ft/min or more and the selected level when the aircraft has sent it (`007↓020`: at 700 ft, descending, 2 000 ft set); the ground speed in tens of knots and the type (`14 A320`), for a departure followed by its runway (`04R↑`). The labels stay beside their squares as the standard ones (traffic to 22R and 04R on the left); the squares are drawn a little smaller. The options Aircraft type on the map and Selected altitude change only the standard labels (the selected altitude still appears in the strips and the selected aircraft's line). There are no extended labels on the phone.

Selected altitude (from 30 October 2026). With the option Selected altitude: On (Off by default) the altitude is followed by the altitude the pilots have set on the autopilot panel (the MCP or FCU), when the aircraft has sent it in the last 2 minutes — from its answers to the radars' Enhanced Surveillance questions (register BDS 4,0) or its own ADS-B. It is written as on the Live Map (from the evening of 30 October 2026): `+` climbing, `–` descending (300 ft/min or more), `=` level, then the selected altitude as `A` and hundreds of feet up to the transition altitude of 5 000 ft and as a flight level above. `A023+A050` is an aircraft at 2 300 ft, climbing, with 5 000 ft set; `FL120–A030` one at FL120 descending, with 3 000 ft set — usually the altitude it has been cleared to; `FL390=FL390` one level at its selected FL390. The selected altitude is as the pilots set it (on the QNH below the transition altitude). Not every aircraft sends it, and the radars ask for it only now and then, so it is missing for some aircraft; it is never guessed. The option Aircraft type on the map (Off by

default) adds the ICAO type after the callsign, e.g. `FIN456 AT75`. With either option on, the map's labels and squares are drawn a little smaller so that the longer labels fit; with both off they are as before.

Traffic to 22R and 04R is labelled to the left, to 22L and 04L to the right, so the two parallel approaches do not overlap. Behind them the ILS centrelines (blue, dashed; the RNP approach to 33 amber), the airport, the tracking range (dashed circle) and, with Trails, each aircraft's path over the last 2 minutes. Aircraft on the map: On the approach (default; those of the chosen runway) or All tracked — also the aircraft near the airport outside the approach corridors, dimmed. *Tap* an aircraft on the approach to follow it, as on the profile. Zoom with two fingers or + −, move with one finger; the map keeps the place and zoom you leave it at, and the house button under them takes you back to the airport. To scroll the page down to the strips, swipe on the profile or the strips, not on the map. The map is light or dark with the theme.

A landed aircraft (from 29 October 2026). After touchdown the receiver still hears an aircraft for a while, so before it stayed in the view for about one and a half minutes at its last position, about 0.6 NM before the threshold. Now an aircraft within 2 NM of the threshold and below 1 000 ft whose position has not changed for 15 seconds is shown *faded*: a grey square with `landed` on the map, a grey dot on the profile and a dimmed strip with `LANDED?` and `no new position for 22 s`. After 30 seconds it is no longer shown, and Auto has already moved the barbs to the next aircraft. Only the display changes; the landing is recorded as before.

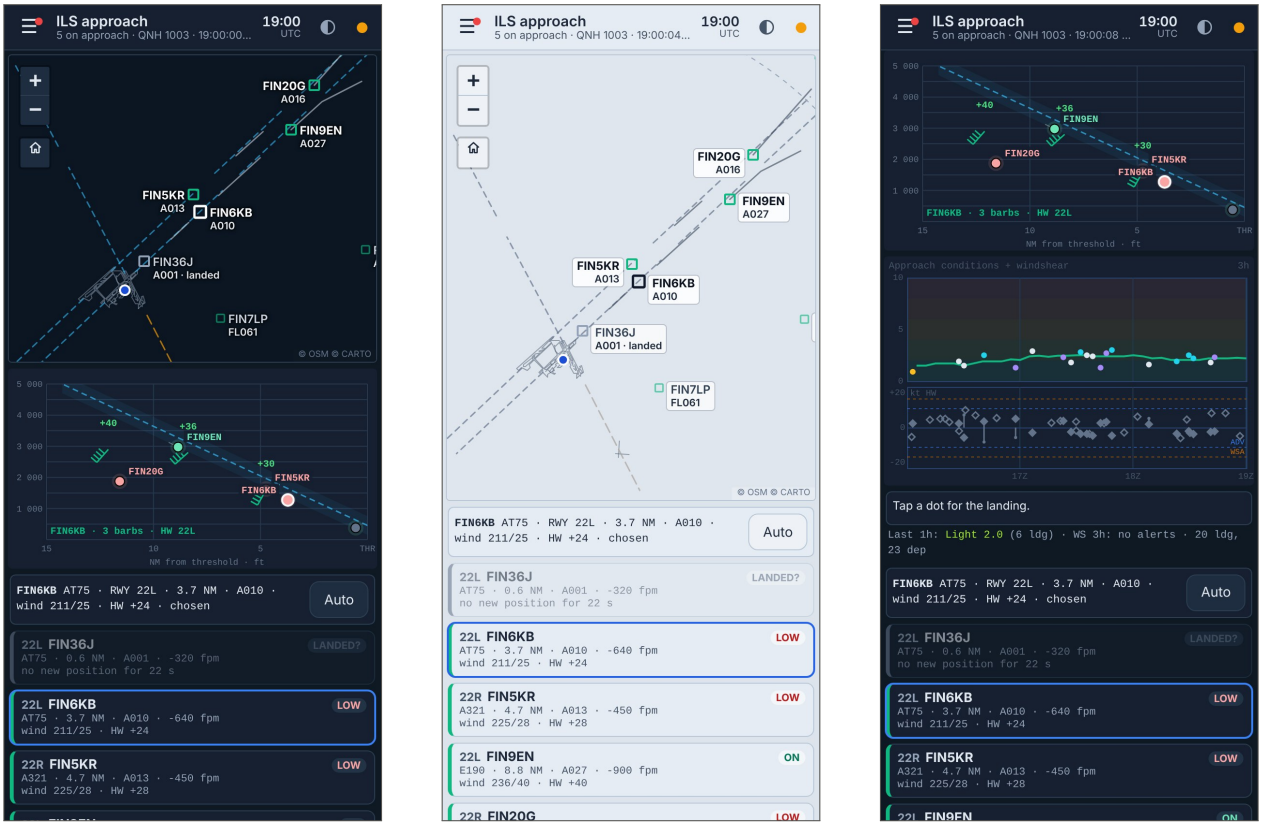


Figure 11. The ILS approach with the map: five aircraft on 22L and 22R (approach traffic replayed from the stored observations of 27 October 2026, with QNH 1003 hPa; FIN36J is shown as just landed; in these pictures the map tiles — streets, coast, lakes — could not be loaded, on the phone they are under the aircraft). Left: Show: Both — the map, the lower profile and the strips; FIN6KB has been tapped, so its square is white and the profile shows its barbs; Auto goes back to the lowest. Middle: Show: Map in the light theme, with room for four strips. Right: the chart WS + ACI (3 h) under the profile, scrolled down; the chart with the stored landings and departures of 28 October 2026 16–19 UTC.

Sizes. In portrait with Show: Both the map takes about 40 % of the screen height and the profile about a quarter, so that the first strips are visible under them; with Map the map takes 58 % and leaves room for about four strips.

Chart under the view (from 29 October 2026; from 30 October on by default). The option Chart under the view — WS + EDR (the default), WS + ACI or Off — adds a chart of the ACI · WS · ST view to the ILS view (Fig. 11, right: WS + ACI): on top the turbulence (EDR, Section 7) of every landing (dot) and departure (ring), or how rough each landing was (the Approach Conditions Index), and below the windshear report of every landing and departure as a change of headwind in kt, for the last 3, 6 or 12 hours (Chart range, 3 h by default). It is under the profile, or under the map with Show: Map, and the strips follow it. Tap a dot for that landing or departure; the line under it gives the last hour's turbulence or conditions and the windshear alerts of the period. Section 7 explains the chart. Its data are fetched at most once a minute.

Whose barbs. By default the barbs follow the *lowest* aircraft on the approach, the one closest to the threshold — the

freshest picture of the wind it is flying through — and change to the next aircraft when it lands. Tap another aircraft on the profile or the map, or its strip, to follow that one; the line under the profile then says `chosen` and an Auto button takes you back to the lowest.

The strips under the profile list the aircraft on the approach, nearest first: runway, callsign (`AP`: autopilot), any windshear alert of that aircraft, the glideslope status; aircraft type, distance, altitude (`A010`, see above) and vertical speed; the latest wind and its headwind (`HW`) or tailwind (`TW`) component. The coloured left edge is the wind source as for the barbs. The top bar gives the count, the QNH used for the glideslope and the active alerts.

Option	Choices	What it does
Show	Profile, Map, Both	the profile, the map, or both (default)
Aircraft on the map	On the approach, All tracked	only the aircraft on the approach (default), or also the others within the tracking range
Departures on the map	Hidden, Shown	the climb-outs, marked with their runway and ↑ (default Hidden)
Turbulence on the map	Off, 30 min, 1 h	the paths of the latest landings and departures coloured by their EDR, and a summary line (default Off; research)
Labels on the map	Standard, ATC	callsign / altitude, or the ATC style's three lines `FIN3YX` / `007↓020` / `14 A320` (default Standard)
Aircraft type on the map	Off, On	the ICAO type after the callsign, e.g. `FIN456 AT75` (default Off)
Selected altitude	Off, On	the altitude set on the autopilot panel after the altitude, e.g. `FL120–A030` (default Off)
Speed vectors	Off, 30 s, 1 min, 2 min	a line to where each aircraft will be after that time on the map (default Off)
Trails	switch	each aircraft's path over the last minute (profile) and 2 minutes (map)
Runway	All, 04L+R, 22L+R, 04L … 33	the approaches shown (profile, strips and map)
Range	15 NM, 7.5 NM	7.5 NM spreads the last part of the approach
Wind barbs	Lo, Hi, Off	Lo: one barb per 400 ft / 0.5 NM; Hi: per 150 ft / 0.2 NM (denser, more scatter)
Wind component	Headwind, Crosswind, Off	the number above each barb
Windshear alerts	ADV +, WSA +, MBA, Off	the lowest alert level drawn (the same setting as on the Windshear page of the same browser)
Chart under the view	WS + EDR, WS + ACI, Off	turbulence (EDR) or approach conditions (ACI) above the windshear of each landing and departure, under the profile or the map (default WS + EDR)
Chart range	3 h, 6 h, 12 h	the period of that chart (default 3 h)

Table 3. ILS approach options. The *Windshear detection and the Windshear page* guide explains the alerts and the barbs.

7 ACI · WS · ST

The approach conditions and windshear chart of the Windshear page (Fig. 12). The upper part is the *Approach Conditions Index* (ACI, 0–10) of every landing: how rough the last part of the approach was, from the aircraft's own replies — below 2 smooth, 2–4 light, 4–6 choppy, 6–8 rough, 8 and above very rough. The dots are coloured by the aircraft class (`N` narrowbody, `R` regional jet, `T` turboprop, `W` widebody), the line is the running median. With ACI + windshear the lower part shows the windshear report of each landing and departure on the same time axis, as a change of headwind in kt: the strongest headwind ramp of a landing as a stem from zero, the change against the previous 45 minutes as a filled diamond (landings) or a hollow one (departures), go-arounds as triangles, and the advisory and alert levels as dashed lines.

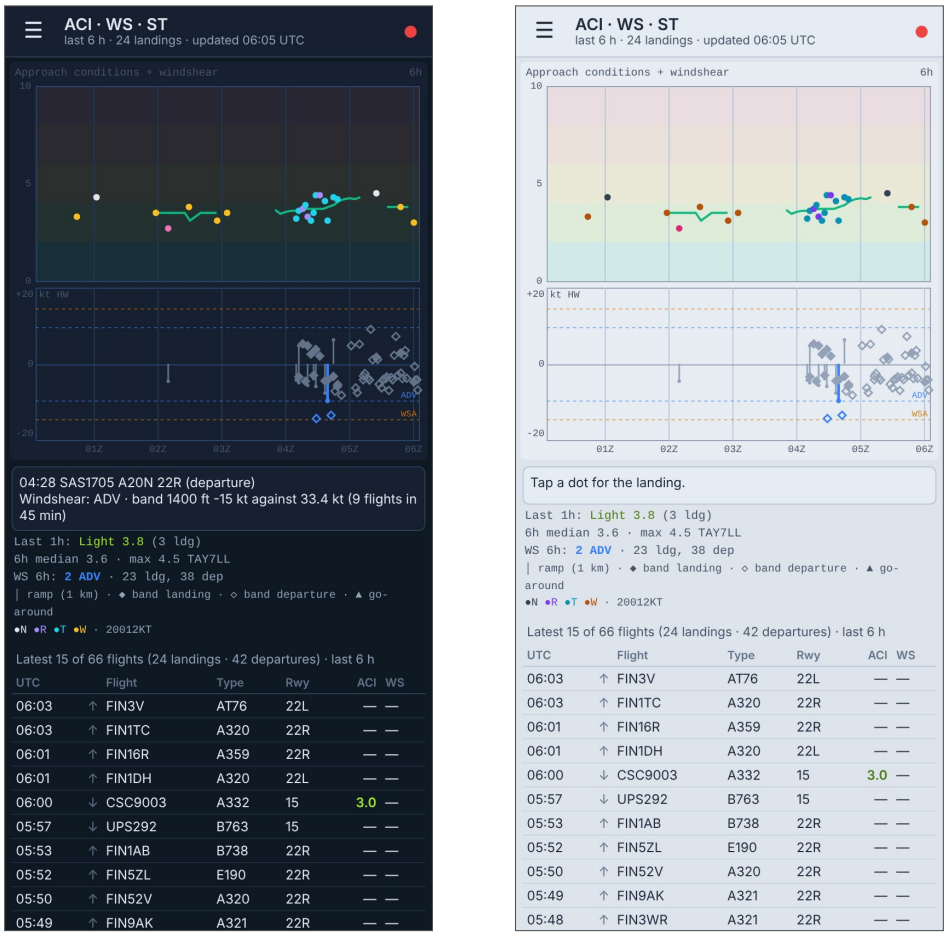


Figure 12. ACI + windshear, the last 6 hours to 28 October 2026 06:05 UTC: landings on runway 15, departures from 22L/22R in a south-westerly low-level jet, ACI mostly 3–4.5. Left: a tap on a diamond shows that flight (here the departure SAS1705, an advisory: 15 kt less headwind at 1 400 ft than the flights before); right: the light theme. The list under the chart is in Fig. 15.

ACI + EDR + WS (from 29 October 2026; research). The fourth Show choice puts the **turbulence** between the two (Fig. 13). For every landing (dot) and departure (ring) MODE-S Wind estimates from the aircraft's own replies the **EDR** — the turbulence measure of ICAO, which calls a peak EDR of 0.10 light, 0.20 moderate and 0.45 severe for a medium-size aircraft. The middle strip has those three lines on a logarithmic scale, the dots coloured green (smooth), yellow, orange or red, and the running median in purple. **Tap a flight** to see its EDR over the whole approach or climb, below and above 1 000 ft and its strongest 2-km piece; a line marks it through all three strips. The summary adds *EDR: last 1h 0.13 light · median 0.12 · max 0.19*. The EDR is an average over the approach from replies about 2 s apart, not the ICAO one-minute peak: a research value, to be compared with the pilot reports. The *Windshear* guide explains how it is computed.

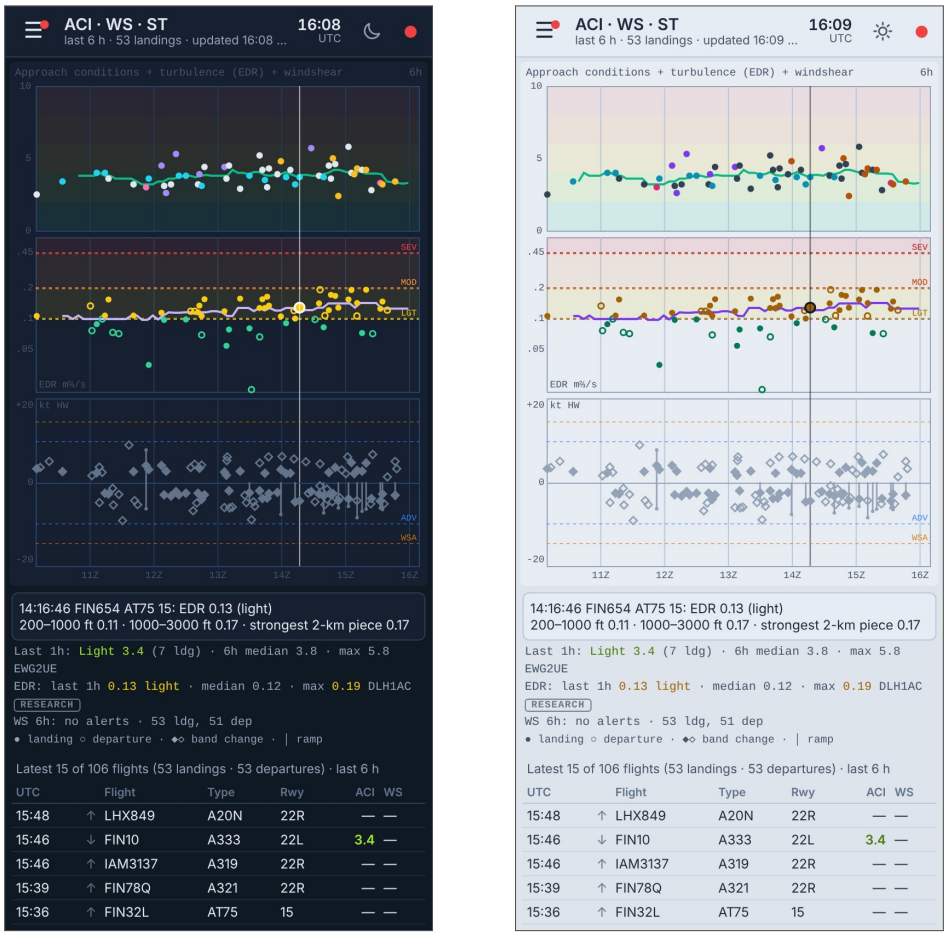


Figure 13. ACI + EDR + WS, 6 hours: the flights of 28 October 2026 07–12 UTC replayed in a test (the time axis shows the replay time). A tapped landing (FIN654, EDR 0.13, light) is marked through the three strips; most landings were light, the departures (rings) a little lower. Right: the light theme.

Boundary layer (from 29 October 2026; research). The choices Boundary layer · wind and · temperature show the lowest 5 000 ft at the airport hour by hour (Fig. 14): the wind speed in colour, or the temperature against the ground (red: warmer than at the ground — an *inversion*, calm and smooth air; blue: colder). A **low-level jet** — a band of strong wind a few hundred metres up, usually at night — is marked by a diamond at its height with its speed, joined hour to hour; a **surface inversion** by a red bar along the bottom. Hatched: no aircraft flew. Tap a cell for its wind or temperature, a diamond for the jet, a red bar for the inversion. The summary gives the last hour, e.g. *low-level jet 750 ft 18 kt from 023° · surface inversion +2 K*. The *Windshear* guide explains how they are found.

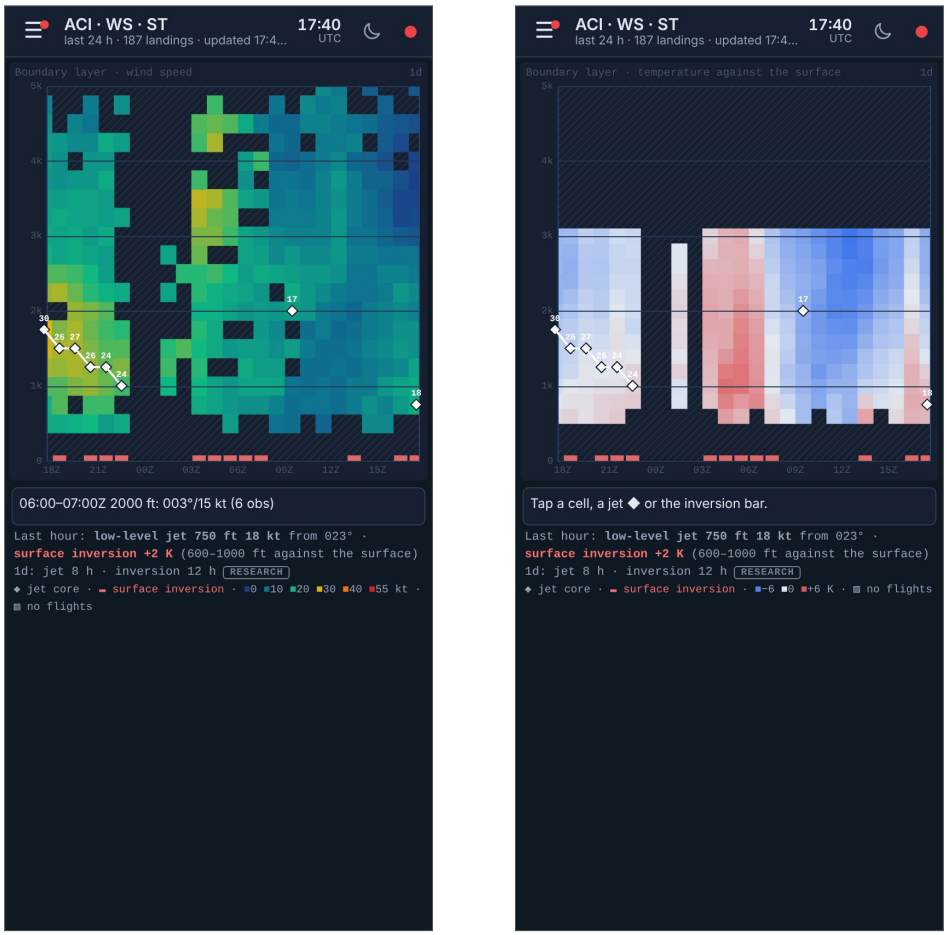


Figure 14. Boundary layer, 17 October 18 UTC – 18 October 17 UTC. Left, wind: the evening jet sinking from 1 800 to 1 000 ft (30 → 24 kt), a tapped cell above. Right, temperature: warmer aloft in the night and morning (red), colder by day (blue); the red bars are the hours with a surface inversion.

Official reports on the chart (from 29 October 2026). When a METAR reports windshear (`WS R22L`) the windshear part has an amber bar along its top from that METAR to the next one, with the runway; a pilot report near the airport (a WXREP or ARS within 30 NM, at or below 10 000 ft) is a teal **P** at the time it was observed (Fig. 20, left). Tap them for the whole message. The summary adds e.g. `METAR WS R22L · 1 pilot report near`. The same marks are on the WS + ACI chart of the ILS view. The messages are listed in the History (Section 8.1).

Under the chart: the box with the tapped landing, departure or go-around; the summary lines of the Windshear page (last hour, median and maximum of the period, the windshear alerts and the counts); and the **list of the flights of the period** (from 28 October 2026; Fig. 15), newest first. Its heading gives the count and the period (*Latest 15 of 59 flights (17 landings · 42 departures) · last 3 h*); 15 rows are shown, Show all opens the rest. The list follows the chart: with ACI only the landings, with ACI + windshear the landings (↓) and departures (↑) — time, flight, aircraft type, runway, ACI (landings only) and windshear level — and with Space–time the flights of the diagram with their largest band change and its height (*−7 @1200*). Landings in History and Departures in History open the History view (Section 8) with the same period, for the bands and reports of each flight. **Options**: the period (3 h, 6 h, 12 h, 1 day) and ACI + windshear, ACI only, ACI + EDR + WS, Space–time or Boundary layer · wind / temperature. The *Windshear* guide and the *Approach Conditions Index* article explain them.

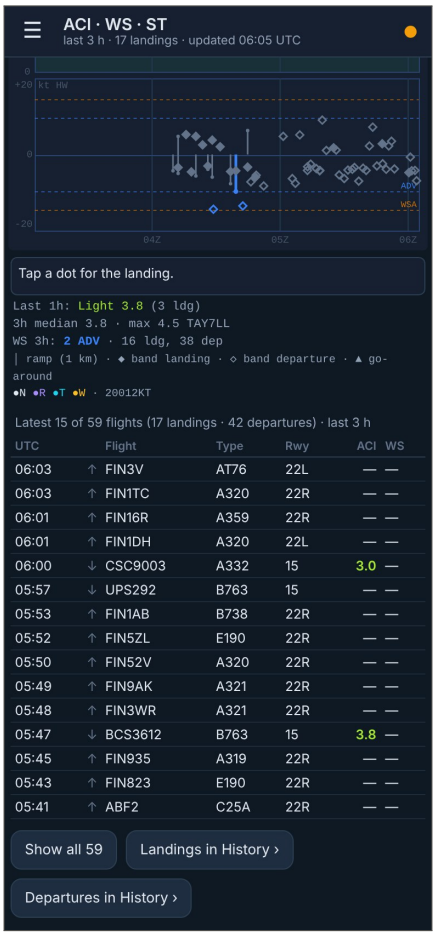


Figure 15. The list under the chart (scrolled down), ACI + windshear, the last 3 hours to 28 October 2026 06:05 UTC: the latest 15 of 59 flights — landings on runway 15 (↓) and departures from 22L/22R (↑) — with Show all 59 and the History buttons.

Space–time (from 28 October 2026; Fig. 16) shows how the wind changes move through the successive flights: time to the right, height (400–1 800 ft) upwards, and in each cell the median change of the headwind of the landings and departures in that 5–30 minutes against the flights of the previous 45 minutes — **blue** less headwind, **red** more, from 3 to 15 kt. A front or a shear layer is a coloured streak through several flights and bands, turbulence scattered single cells. A dot marks a flight at or beyond its advisory limit, a triangle at the top a flight with a windshear level. Tap a cell: the box below shows the cell, and the list its flights with their change and wind (All flights × goes back to the whole period). Turn the phone for shorter time cells.

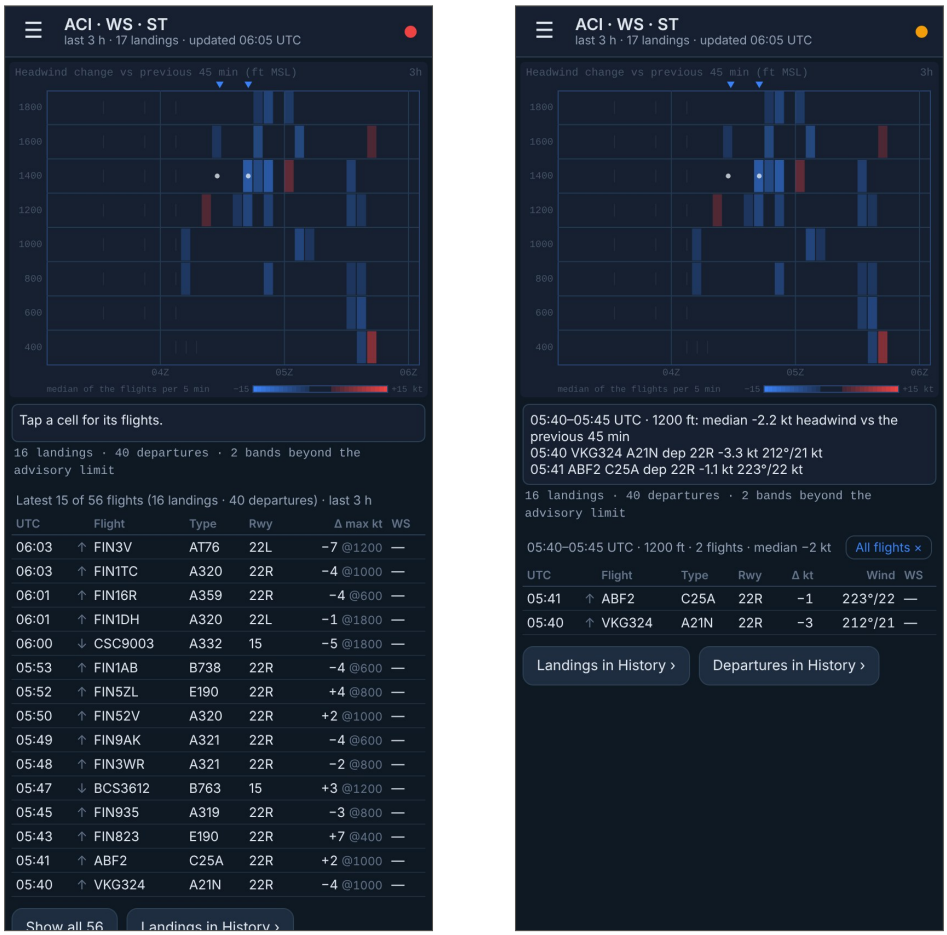


Figure 16. Space–time, the last 3 hours to 28 October 2026 06:05 UTC: the morning departures from 22L/22R climbing into a weakening south-westerly low-level jet — blue cells (less headwind) at 1 200–1 600 ft. Left: the diagram and its flights, each with its largest band change; right: a tapped cell (05:40–05:45 UTC, 1 200 ft) and its two departures in the list.

8 History

The Approach History of the Windshear page made for a phone (Fig. 17): the **landings** or the **departures**, newest first, as cards.

- **Line 1**: time (UTC; earlier days with the date), callsign, aircraft type, runway (in a thin frame; the columns line up from card to card), and chips: the **ACI** value, the **windshear level** (`ADV`, `WSA`, `MBA`), ***trial*** (a departure whose headwind ramps along the climb would have given that level — a trial, no alert), **EDR** (from 29 October 2026: the turbulence of the flight, outlined in its colour — green smooth, yellow light from 0.10, orange moderate from 0.20, red severe from 0.45) and **GA** (go-around).
- **Line 2**: depends on Show. Headwind and Crosswind: one cell per height band (the heights are in the row under the search field) with the component along the runway in kt — headwind green, tailwind red; crosswind + from the right, − from the left. Wind: an arrow (where the wind blows to) and the speed. ACI: the index and its two segments with the METAR. Windshear: the strongest headwind ramp and the band change. Turbulence (from 29 October 2026): the EDR with its class, below (lo) and above 1 000 ft (hi), the roughest 2-km piece (peak) and the number of pieces (Fig. 18).
- **Tap a card** to open it: all its height bands with wind, headwind and crosswind, the approach conditions in detail, the windshear report and the turbulence in words — the texts the full site shows when the mouse is over a row — and, from 29 October 2026, the **boundary layer** of that hour: *Boundary layer then: low-level jet 1750 ft 27 kt from 134°* (or a surface inversion, or neither). Tap again to close.

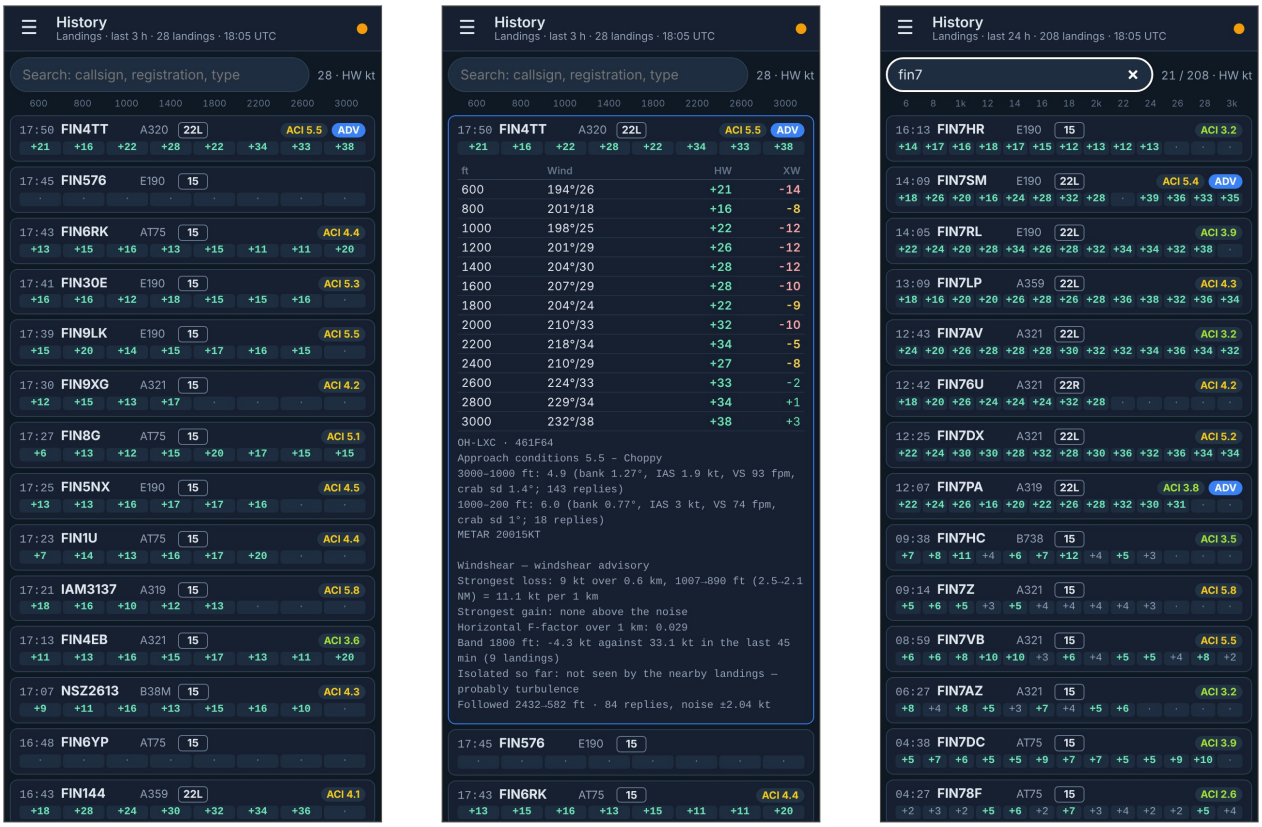


Figure 17. History, 27 October 2026. Left: the landings of the last 3 hours, headwind per band (a strong south-westerly: +13 to +38 kt). Middle: a card opened — the bands, the approach conditions and the windshear report. Right: the search `fin7` over a day, with the 13 height bands of Hi.

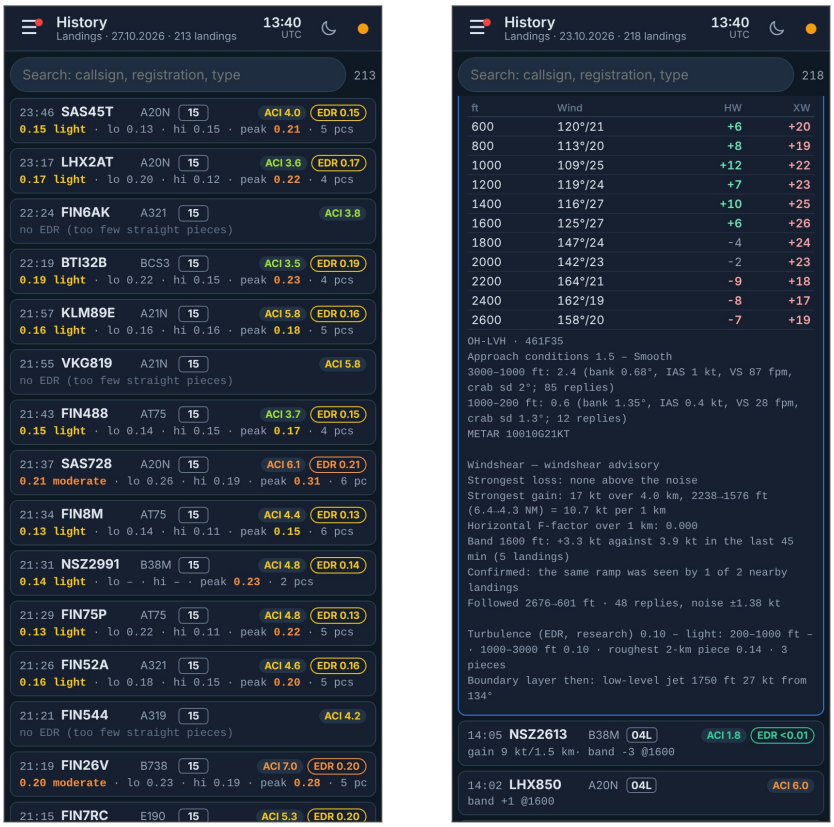


Figure 18. Left: Show: Turbulence, the landings of 27 October 2026 — most light, SAS728 and FIN26V moderate; “no EDR” when the approach had too few straight pieces. Right: an opened card of 23 October with the turbulence and the boundary layer then — FIN440 had a headwind-gain advisory while a low-level jet blew at 1
,750
,ft. The EDR is a research value: an average over the approach, not the ICAO one-minute peak.

Search. The field above the cards finds the callsign, registration or aircraft type anywhere in the text, upper or lower case and with or without the hyphen: `fin7`, `OH-LX` or `ohlx`, `AT75`. The number beside it shows how many of the flights match. The search on the full site’s Approach History works the same way.

Options (Fig. 19): List Landings / Departures (or METAR & reports, Section 8.1); the range 1 h – 1 day (default 3 h); or a **date**: the calendar field or ⟨ ⟩ (a day back / forward), and Live returns to the range; Show; and Lo · 8 or Hi · 13 height bands. A live range is updated every minute; a date is loaded once.

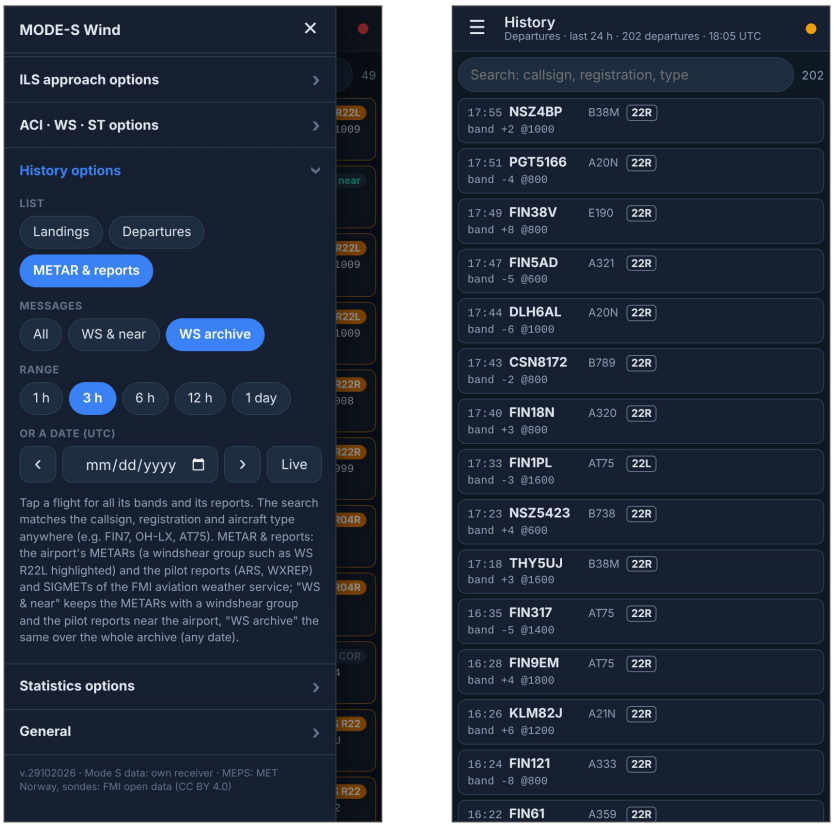


Figure 19. Left: the History options (here with METAR & reports chosen, which has Messages in place of Show and the height bands). Right: the departures of a day, Windshear shown — the band change of each climb-out against the earlier flights.

8.1 METAR & reports

From 29 October 2026 the History's List has a third choice, METAR & reports (Fig. 20): the METARs of Helsinki-Vantaa and the pilot reports and SIGMETs of the FMI aviation weather service, newest first, one card each — the time, the kind (`METAR`; `WXREP` teal, `ARS` red, `SIGMET` violet), for a pilot report the aircraft, what was reported and the height, the chips `WS R22L` (a METAR with a windshear group) and `near`, and the whole message with the windshear group in amber and the reported phenomena (`MOD WS`, `FBL TURB`, `INV` ...) in teal.

- **Windshear in a METAR**: in Finland the group `WS R22L` or `WS ALL RWY` is only in the METARs made by an observer — windshear that an aircraft or the tower reported on a landing or take-off below 1 600 ft. The automatic METARs of the night (`AUTO`) never have it.
- **WXREP** is a pilot report below the ARS criteria: light turbulence, light icing, light windshear, an inversion, the top of a CB. **ARS** is for moderate or severe turbulence or icing and the like.

Options: Messages All, WS & near (only the METARs with a windshear group and the pilot reports near the airport) or WS archive (the same over every stored date — also years back once the METAR history has been loaded); the range or a date as for the flights. The search finds text in the messages: `ws`, `r22l`, `turb`, a callsign.

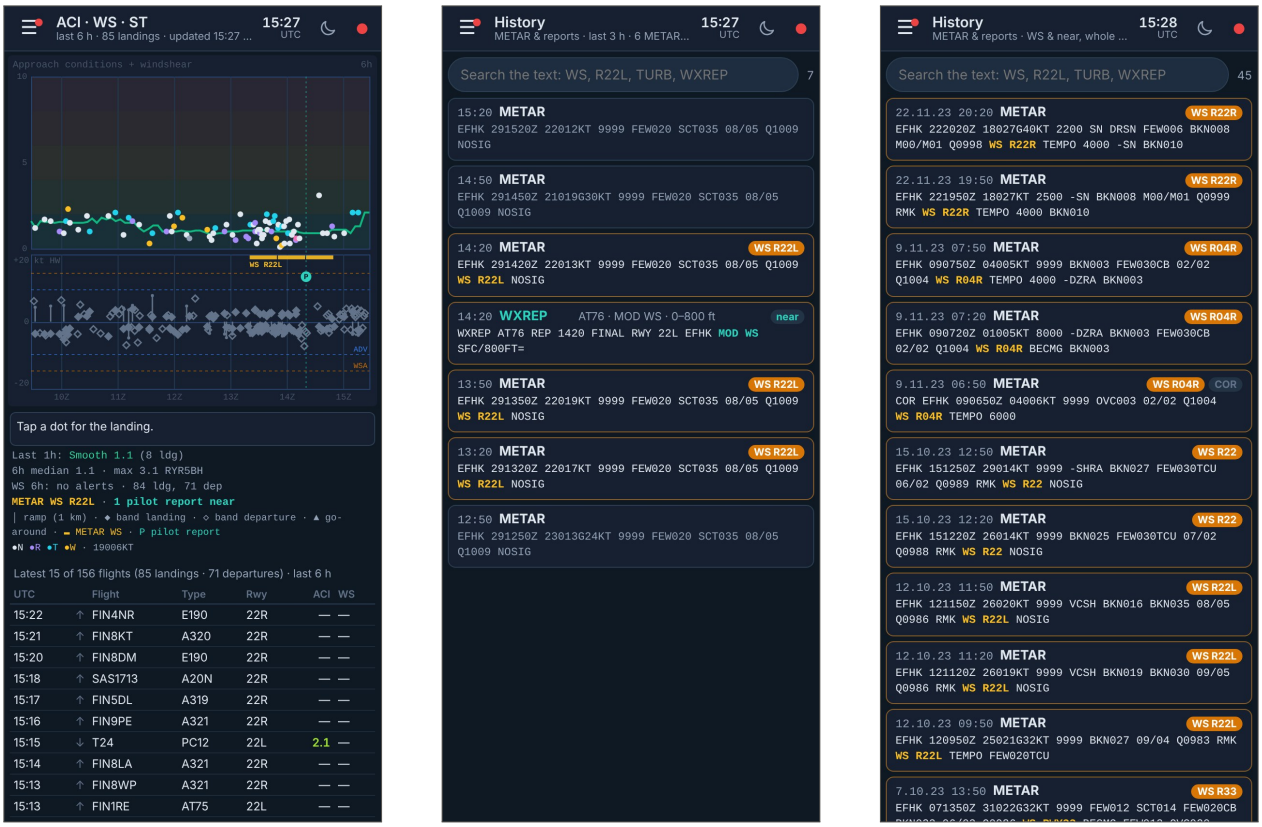


Figure 20. Left and middle: an example drawn with test messages — a METAR `WS R22L` for three half-hours and a WXREP `MOD WS SFC/800FT` on final 22L, on the ACI + windshear chart (amber bar, **P**) and in the History's METAR & reports. Right: WS archive — the real METARs of Helsinki-Vantaa with a windshear group in 2023, e.g. 22 November in snow and a southerly gale, `WS R22R`.

9 Statistics

The Statistics page of the full site made for a phone (from 28 October 2026): what the receiver hears, one panel under the other (Fig. 21). At the top six figures — messages per second, aircraft, aircraft with ADS-B and with EHS (the Comm-B registers that give wind and temperature), the MLAT and ADS-B positions and the age of the receiver's feed — and the time of the last update with the window. Below them the sections you have chosen, by default:

- **Message rate** — the messages per second as a chart (1 h, 24 h, 7 d above it; the selector chooses the message class). Its second tab, Data transfer, shows how much data the server receives from the receiver and sends to the users.
- **Downlink formats** — the share of each kind of Mode S reply, e.g. DF17 ADS-B, DF20/21 the Comm-B replies that carry the meteo registers, DF11 the all-call replies.
- **Meteo yield** — how many winds and temperatures per minute the replies give, and why the rest gave none.
- **Range rings** — how far the receiver hears aircraft in each direction, for three height layers, with a table per distance. *Tap* the rings for the farthest aircraft in that direction and its height.

Each panel's explanation is folded under About this panel; tap it to read it. The panels are the same as on the full site, so the *Mode S & ADS-B and the Statistics page* guide explains them in full.

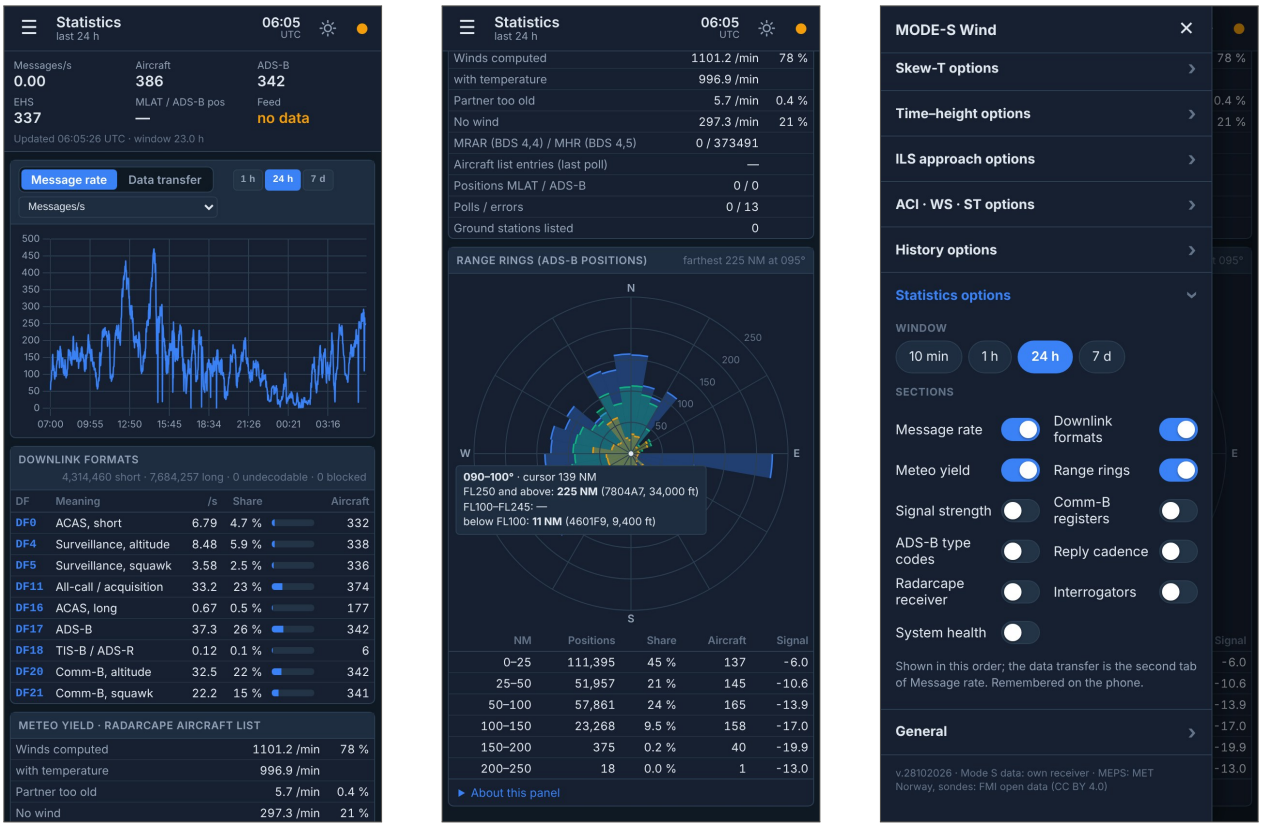


Figure 21. Statistics, last 24 h (a copy of the database of 28 October 2026 without a live feed, so the top figures show 0 messages per second and *no data*). Left: the summary, the message rate and the downlink formats. Middle: the range rings with the read-out of a tap towards the east. Right: the Statistics options — window and sections.

Options (Fig. 21, right): the **window** 10 min, 1 h, 24 h (default) or 7 d — the period of the tables, the downlink formats, the signal histogram and the range rings; and a switch for each **section**. Besides the four above you can switch on Signal strength (the histogram of the reply levels), Comm-B registers, ADS-B type codes, Reply cadence (how often each aircraft is interrogated), Radarcape receiver (its GPS, timestamps and temperature), Interrogators (the radars and WAM stations that ask, from the all-call replies) and System health (the indicators of the health dot with their details). The sections are always shown in this order, and the choice is remembered on the phone.

Updates and data. The view is updated every 30 seconds while it is shown (the full site's page every 10 seconds), and not at all while another view is shown. Each update is about 55 kB, so an hour of watching uses about 7 megabytes of mobile data — more than the other views; switch to another view when you have seen what you wanted. The chart program and the Statistics code are loaded the first time you open the view.

10 When something looks wrong

- **The page does not load, or asks for the password again and again.** The phone must reach MODE-S Wind as a computer does; check the network and the address (with `/app`). The login is remembered for 30 days; log in again once after your password has been changed, or when the phone's browser data has been cleared.
- **`Could not load ...`** under a view: the server did not answer in time. The view tries again at its next update, or switch the view and back.
- **The health dot is amber or red.** Open the menu: General says which part. MODE-S Wind keeps working with what it has — e.g. without new MEPS data the Skew-T has no model line, the rest is unchanged.
- **`No aircraft on the approach`**: there is no traffic in the approach corridors of the chosen runway; choose All.
- **`MEPS: no model data for this time`** or `no sounding stored`: the model run or the radiosonde for that time has not arrived (or is not available); the aircraft profile is not affected.
- **After an update of MODE-S Wind** the phone may still show the old page from its cache: reload the page in the browser, or close the home-screen app and open it again.
- **Text too small?** Use landscape, or the phone's own zoom (pinch); the diagrams are drawn at the phone's full resolution.

11 Quick reference

Item	Meaning
`/app`	the mobile view; same login as the full site, remembered for 30 days
≡	menu: views as tiles, options of each product, General
Red dot on ≡	a new version: open What's new at the top of the menu (or in General)
Time at the top	the current time in UTC (Finland: + 2 h in winter, + 3 h in summer)
Moon / sun / half circle	theme Dark / Light / Auto (follows the phone); a tap steps to the next
Dot at the top right	system health: green working, amber late / fallback, red stopped
Tap / drag	values at that point (the mouse hover of the full site)
Swipe up / down	scroll a view taller than the screen
Grey line and barbs	aircraft profile (Skew-T), with the wind values
Pink / teal	Jokioinen radiosonde / MEPS weather model
Green / amber / red dot (ILS)	aircraft on / above / below the glideslope (±300 ft)
Squares on the map (ILS)	aircraft; colour = wind source, framed = whose barbs are drawn; tap to follow
`ADV` / `WSA` / `MBA`	windshear advisory / windshear alert / microburst alert
`HW` / `TW` / `AP`	headwind / tailwind component; autopilot engaged
ACI	Approach Conditions Index 0–10: below 2 smooth, light, choppy, rough, 8+ very rough
Updates	Skew-T now 2 min, time–height 10 min, ILS 5 s, ACI · WS · ST and History 1 min, Statistics 30 s; only the shown view, only with the screen on
Blue / red cells (ST)	less / more headwind than the flights of the previous 45 min
Card chips (History)	ACI value, windshear level, *trial* (departure ramps, no alert), GA go-around; for the reports `WS R22L` and `near`
Amber bar / **P** (WS chart)	a METAR with a windshear group / a pilot report near the airport
About this panel (Statistics)	the panel's explanation, folded; tap to open
Settings	remembered on the phone; Skew-T options and theme shared with the full site in the same browser

Other guides (the User guides tab of the full site's *What's new* window): *Reading the Atmosphere Profile (Skew-T)*; *The Sounding and Wind Map pages*; *Windshear detection and the Windshear page*; *Mode S & ADS-B and the Statistics page*.